

Ankit Kumar

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2025 - Present	M.Tech (DoMS)	Indian Institute of Technology, Kanpur	7.18/10
2021 - 2025	B.Tech (Agricultural Engineering)	C.S.A. University of Agriculture and Technology, Kanpur	8.25/10
2021	UP Board (XII)	Government Inter College Kadaura, Jalaun	77.20%
2019	UP Board (X)	Government Inter College Kadaura, Jalaun	80.83%

Professional Experience

- Tata Consultancy Services (TCS) | Data Science & AI Intern | Computer Vision & ML** (May'25 - Jul'20)
- Engineered an automated end-to-end Vehicle CO2 Emission Tracking & Intelligence pipeline from traffic CCTV feeds to monitor urban carbon footprints and support enterprise ESG sustainability compliance.
 - Implemented YOLOv8 deep learning localization with an OpenCV preprocessing engine (Bilateral Filtering, CLAHE, Otsu Binarization) and EasyOCR, paired with a custom Indian plate regex syntax corrector (resolving optical confusions like O vs 0, I vs 1, B vs 8) to enhance OCR character recognition on degraded video frames.
 - Integrated Vahan / RTO registry APIs & offline databases across 36 Indian States/UTs to dynamically extract vehicle metadata (Fuel Type, Engine CC, Vehicle Class) and compute dynamic Vehicle Age (Current Year - Manufacturing Year).
 - Formulated a Multi-Origin Convergence Routing algorithm using Haversine & OSRM distance matrices to model multi-vehicle trip trajectories and calculate fleet-level carbon density.
 - Trained dual-stage ensemble ML models, where Gradient Boosting Regressor achieved $R^2 = 0.996$ (RMSE = 3.12 g/km) for emission rate prediction and Random Forest Regressor achieved $R^2 = 0.989$ for total trip emissions (kg), benchmarked against ARAI/IPCC emission standards.
 - Delivered an interactive Streamlit Web Dashboard & automated multi-sheet Excel reporting pipeline, featuring live bounding box annotations, GIS convergence maps, and a 17-test Pytest validation suite with 100% pass rate.

Key Projects

- Zepto E-Commerce Data Analysis | SQL | Exploratory Data Analysis** (Self Project, Aug'25 - Sep'25)
- Analyzed 3,700+ Zepto product records across 14 categories, identifying pricing, discount, and inventory trends.
 - Performed data cleaning by removing invalid zero-price records and converting paise to INR for accurate analysis.
 - Applied optimized SQL queries using SELECT, WHERE, GROUP BY, HAVING, CASE, and aggregations for business analysis.
 - Generated insights on discounts, stockouts, revenue, inventory weight, and price-per-gram to support pricing decisions.
- Customer Shopping Behavior Analysis | SQL | Power BI** (Self Project, Sep'25 - Oct'25)
- Analyzed 3,900 customer records with 18 attributes using Python and Pandas to identify retail behavior patterns.
 - Cleaned 37 missing ratings and engineered age groups, spending tiers, and purchase frequency features for segmentation.
 - Applied PostgreSQL queries using CTEs, window functions, subqueries, CASE, and aggregations for business analysis.
 - Built an interactive Power BI dashboard with DAX measures, KPIs, slicers, and drill-throughs for customer insights.
- Loan Portfolio Risk Analysis & Visualization | SQL | Tableau** (Self Project, Aug'25)
- Analyzed 38.58K loan applications to evaluate funding amounts, repayments, interest rates, DTI, and portfolio risk.
 - Applied SQL queries using GROUP BY, CASE, aggregations, and filtering to segment loan portfolios by loan status.
 - Identified 86.18% good loans and 13.82% bad loans, delivering critical insights to support credit risk assessment.
 - Built interactive Tableau dashboards tracking loan applications, funded amounts, and monthly performance trends.
- Credit Risk Modelling: Loan Eligibility Prediction | Classification** (Guide: Prof. Faiz Hamid, Aug'25 - Nov'25)
- Built an end-to-end credit risk model using 51,336 customer records and 79 features to predict loan eligibility.
 - Performed feature engineering and statistical hypothesis testing using Chi-Square, ANOVA, VIF, and SMOTE for data.
 - Developed robust ML pipelines using Logistic Regression, Decision Tree, SVC, Random Forest, and XGBoost classifiers.
 - Optimized XGBoost with GridSearchCV tuning, achieving 78% accuracy for loan eligibility prediction on test data.
- BigMart Sales Analysis Using Multivariate Regression | Causal Analysis** (Self Project, Aug'25 - Sep'25)
- Analyzed 8,523 BigMart sales records using Pandas, Seaborn, and Matplotlib to identify key product sales drivers.
 - Performed data cleaning and feature engineering using grouped imputation, encoding, and outlet-age features for modeling.
 - Applied OLS and Linear Regression with log transformation and VIF analysis, achieving a solid 72.1% R^2 score on data.
 - Selected four key predictors including Item MRP and outlet types, achieving 0.290 cross-validation MSE on sales data.
- AcademicPath: Career & Course Guidance System | Multimodal RAG** (Self Project, Oct'25 - Nov'25)
- Developed a multimodal RAG assistant using LlamaIndex, Qdrant, Hugging Face embeddings, and Groq LLMs for guidance.
 - Implemented semantic vector retrieval with course metadata and page citations across 113+ academic college courses.
 - Built course-comparison routing and conversational memory to generate structured guidance for multi-course queries.
 - Integrated Faster-Whisper and gTTS for real-time voice interaction, achieving 100% Context Recall and 70.1% Accuracy.
- MediMind AI: Clinical Decision Support Platform | GenAI | RAG | LLMOps** (Self Project, Nov'25 - Dec'25)
- Built an end-to-end RAG platform using LangChain, Groq API, and Pinecone, reducing query inference latency by 85%.
 - Implemented MiniLM embeddings and Cosine Similarity, achieving 96.8% RAG faithfulness with clear source citations.
 - Developed lab report analysis using PyPDF, detecting biomarker anomalies and exporting structured PDF clinical briefs.
 - Integrated 4 risk calculators and Plotly visualizers with sub-10ms emergency triage guardrails for clinical decisions.

Dental Appointment Management System | Agentic AI | LangGraph (Self Project, Dec'25 - Jan'26)

- Developed conversational AI for dental appointment management, enabling **natural-language booking and rescheduling**.
- Implemented **LLM tool calling** with **LangGraph** and **LangChain** for live appointment search and MySQL database updates.
- Integrated structured appointment **dataset of 4,280 slots across 10 doctors and 7 specializations** for dental clinics.
- Built an **interactive LangGraph workflow** with state management and conditional tool execution for dental bookings.

Text-to-SQL & Agentic SQL Analytics System | LangGraph (Self Project, Jan'26 - Feb'26)

- Developed a **Text-to-SQL** system using **Gemini 2.0 Flash**, **LangChain**, and **MySQL** for natural-language SQL query tasks.
- Built an **agentic LangGraph** ReAct workflow using **SQLDatabaseToolkit** for schema inspection, query generation, and execution.
- Integrated **SQL query-checking tools** to iteratively validate queries and generate answers across enterprise data.
- Evaluated the pipeline using **RAGAS**, achieving **1.00 Context Precision** and **3.8/5 Helpfulness** across benchmark queries.

Air India Domain-Specific GenAI Knowledge Assistant | RAG | LLM (Self Project, Nov'25 - Dec'25)

- Built a domain-specific **AI assistant** using **LangChain**, **Groq Llama 3.3 70B**, and **Streamlit** for Air India document Q&A.
- Processed 121 PDF pages using **DirectoryLoader** and **PyPDFLoader**, with cached loading for fast application startup.
- Implemented **context-grounded prompting** with strict anti-hallucination guardrails, restricting answers to PDF docs.
- Integrated **dynamic LLM routing** between **Groq** and **xAI**, with **real-time token streaming** and user message chat history.

Solar Panel Defect Classification | Deep Learning | Computer Vision (Guide: Prof. Veena Bansal, Jan'25 - May'25)

- Developed a 6-class **CNN** system using **TensorFlow** and **transfer learning** to classify 885 solar-panel field image sets.
- Applied image **augmentation**, **Batch Normalization**, **Dropout**, and **L2 regularization** for superior model generalization.
- Benchmarked **CNN**, **MobileNetV2**, and **EfficientNetB0**, achieving **82.5% validation accuracy** after Keras Tuner optimization.
- Deployed optimized model using **Streamlit**, providing confidence scores, top-3 predictions, and probability plots.

Intelligent Document Processing for Invoice Extraction | OCR (Self Project, Dec'25 - Jan'26)

- Built an invoice processing pipeline using **Python**, **pdfplumber**, **PyMuPDF**, **Tesseract OCR**, and **OpenCV** for PDFs data.
- Processed 1,007 invoice PDFs with 100% pipeline success, extracting structured fields, tables, and invoice line-items.
- Implemented validation and confidence scoring, achieving **98.6% pass rate** and **0.948 average field confidence score**.
- Developed **FastAPI** and **Streamlit** apps processing invoices in **0.13 seconds**, generating JSON, CSV, and RAG outputs.

Marketing Campaign Performance Analysis | Statistical Testing | Regression (Self Project, Sep'25 - Oct'25)

- Evaluated Facebook and AdWords campaigns using **EDA** to compare clicks, conversions, costs, and marketing performance.
- Performed **A/B testing** using Welch's t-test, proving Facebook conversions were significantly higher than AdWords.
- Built **Linear Regression** to model click-to-conversion relationships, achieving a strong **R2 score of 76.35%** overall.
- Applied **cointegration testing**, confirming significant long-term equilibrium between digital ad costs and conversions.

Sentiment Analysis of Amazon Product Reviews Using Hybrid AI Techniques (Self Project, Oct'25 - Nov'25)

- Analyzed 11,000 Amazon reviews using **NLTK**, **tokenization**, **lemmatization**, and **TF-IDF** for product sentiment classification.
- Compared **Logistic Regression**, **Random Forest**, and **Naive Bayes**, with Logistic Regression achieving **85.4% accuracy**.
- Developed **RNN**, **LSTM**, **Bi-RNN**, and **Bi-LSTM** models using **Word2Vec**, with LSTM achieving **85.6% accuracy** on review text.
- **Fine-tuned BERT** for sentiment classification, achieving **90.4% accuracy** and **90% F1-score**, outperforming baselines.

Multi-Agent Resume Analyzer & Job Matcher | GenAI | CrewAI (Self Project, Jan'26 - Feb'26)

- Built a multi-agent resume analyzer using **LLMs**, **CrewAI** architecture, and **sequential workflows** for resume evaluation.
- Developed market research and SWOT agents using **Serper API** and **web scraping** to identify role-specific industry skills.
- Implemented resume-job matching using **TF-IDF**, **cosine similarity**, and **technical keyword extraction** for candidate skills.
- Built a **Streamlit** dashboard displaying match scores, skill gaps, SWOT analysis, and actionable career recommendations.

Electricity Demand Forecasting | Time Series | Deep Learning (Self Project, Aug'25 - Oct'25)

- Forecasted **monthly electricity demand** using statistical and deep learning approaches to support utility planning.
- Conducted time series **decomposition**; tested stationarity using **ADF test** and first-order series data differencing.
- Applied **ARIMA**, **SARIMA**, **Prophet**, and **LSTM** models using **ACF/PACF analysis** to capture seasonal demand trend patterns.
- Achieved the lowest **MAPE (2.46%)** on test data using **LSTM**, forecasting electricity demand for upcoming **36 months**.

Petrol Price Forecasting & Analytics System | Time Series | Machine Learning (Guide: Prof. Faiz Hamid, Aug'25 - Nov'25)

- Built a time-series forecasting system using **ARIMA**, **XGBoost**, **PyTorch LSTM**, and **Naive Baseline** models for fuel prices.
- Engineered 15+ temporal features including **lags**, **SMA**, **EMA**, **volatility**, **returns**, and **calendar variables** for fuel prices.
- Benchmarked models on 35% unseen data, with **XGBoost** achieving **1.12% MAPE** and 67.99% directional movement accuracy.
- Developed **Streamlit** dashboard and **FastAPI** REST API with Plotly plots and model comparison suites for forecasting.

Speech Sentiment Analysis using RNN & LSTM | NLP | Deep Learning (Guide: Prof. Veena Bansal, Jan'25 - May'25)

- Developed a Speech-to-Sentiment pipeline using **SpeechRecognition**, **GloVe**, and **Keras LSTM** for audio classification.
- Processed 14,402 airline tweets using text preprocessing and 100-dimensional GloVe embeddings for model training.
- Built a 2-layer **LSTM** network with 128 hidden units and **Dropout**, achieving **77.20% test accuracy** on airline tweets.
- Integrated **SoundDevice** recording and speech-to-text conversion for real-time sentiment prediction across 3 classes.

McDonald's Customer Segmentation & Marketing Analytics | Clustering | Machine Learning (Self Project, Sep'25 - Nov'25)

- Analyzed 1,453 customer survey records across 11 perception attributes to identify behavioral segments and personas.
- Benchmarked **K-Means**, **K-Modes**, and **Latent Class Analysis** using **StandardScaler**, **PCA**, and **correlation analysis** methods.
- Optimized K-Means with 100-fold bootstrap validation, achieving **0.592 Silhouette Score** and **0.930 Mean ARI** at K=5.
- Profiled five customer personas using demographics and behavior, translating segments into marketing strategies.

Sentiment-Based Product Recommendation System | NLP | Recommender (Self Project, Dec'25 - Jan'26)

- Analyzed 30K product reviews using **NLP preprocessing**, **lemmatization**, **TF-IDF**, and **SMOTE** for balanced classification.
- Benchmarked **Logistic Regression**, **Random Forest**, and **XGBoost**, with **Random Forest** achieving **91% classification accuracy**.
- Developed User-User and Item-Item collaborative filtering using adjusted **cosine similarity**, achieving 2.55 RMSE.
- Built a hybrid recommendation pipeline combining Top-20 candidates and sentiment reranking for personalized Top-5 outputs.

NVIDIA Stock Price Prediction | Time Series Analysis | Quantitative Analysis

(Self Project, Dec'25 - Jan'26)

- Developed forecasting models for Nvidia stock prices using 15 years of data with statistical and ML methodologies.
- Conducted data visualization and examined stationarity, seasonality and trends using ADF, ACF and PACF plots daily.
- Implemented statistical models: ARMA, ARIMA, and SARIMA, along with Deep Learning (LSTM) for stock price sequences.
- LSTM outperformed all models with RMSE = 11.4, followed by SARIMA (RMSE = 21.3) on out-of-sample stock market data.

AI-Powered Gen Z Consumer Behavior & Marketing Intelligence | MR project

(Guide: Prof. Giridhar, March'26 - May'26)

- Analyzed 5,203 consumer records, engineering 41 behavioral and demographic features for consumer purchasing analysis.
- Applied Chi-Square and Cramer's V, identifying significant FOMO and income-spending relationships with $p < 0.0001$.
- Benchmarked 6 classification models using Stratified 5-Fold CV, with tuned CatBoost achieving 67.5% accuracy rate.
- Applied TreeSHAP and K-Means segmentation, identifying key purchase drivers and 4 distinct consumer persona profiles.

Data-Driven Supply Chain Solution | Supply Chain Analytics | ML

(Self Project, Feb'26)

- Analyzed diverse supply chain data for 100 SKUs, effectively identifying inefficiencies in product and transport route.
- Performed dynamic EOQ optimization, thoroughly assessed potential risks, and evaluated inventory KPIs for warehouse.
- Applied robust LightGBM (37% RMSE) for sales forecasting and optimized KNN algorithms for vehicle route delivery.
- Delivered data-driven insights improving stock levels, reducing transport cost, and boosting forecasting accuracy.

Cancer Risk Level Prediction | XGBoost | Multiclass Classification

(Self Project, july'26-Aug'26)

- Analyzed 2,000 patient records across demographic, lifestyle, environmental, and genetic factors for multiclass risk classification.
- Identified and removed target leakage from Overall Risk Score,addressed class imbalance using SMOTE and class-weighted learning
- Optimized XGBoost with Optuna using 40 trials and 3-fold stratified cross-validation for multiclass risk prediction..
- Achieved 88 % test accuracy, 0.72 Macro-F1, and 0.87 weighted F1, deploying the prediction system through Streamlit.

Coursework & Skills

Courses	Probability & Statistics Data Mining and Knowledge Discovery Introduction to Computing Operations Research for Management Applied Machine Learning (AML) Statistical Modelling for Business Analytics(SMBA) Business Process Management Supply Chain Management
Online Courses	Python for Beginners – Udemy Python Machine Learning: From Beginner to Pro – Udemy Power BI Workshop – Office Master Geodata Processing using Python and Machine Learning – IIRS (ISRO) Deloitte Australia Data Analytics Job Simulation – Forage Oracle Cloud Infrastructure 2025 Certified Data Science Professional – Oracle University Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – Oracle University
Technical Skills	Python, SQL, Excel, Power BI, Tableau Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch GenAI, LangChain,Ollama, LangGraph, LlamaIndex, Qdrant, Pinecone, Streamlit, FastAPI, Github, Autoencoders, Transfer Learning,
Soft Skills	Team Management, Leadership, Decision Making, Communication Skills, Adaptability, Teamwork, Problem Solving, Critical Thinking

Positions of Responsibility

- Serving as Alumni and Corporate Relations Coordinator at IITK: Organized 10+ alumni webinars, boosting networking opportunities.
- Served as a Campus Coordinator for Mimamsa 2025 conducted by IISER Pune.

Scholastic Achievements & Extra-Curricular Activities

- Secured 98.48 percentile in GATE 2025 conducted by IIT Roorkee in Agricultural Engineering .
- Awarded the Silver Medal Award for outstanding academic performance in B.Tech by the Chancellor
- Volunteered with NSS for 2 years,organized Girls Education, literacy and poverty survey campaigns under Ek Bharat Shreshtha Bharat.